

JUAN SEBASTIAN PEÑA DONNEYS

Madrid, Spain

(+34) 697 643 097 · juansebastianpenadonneys@gmail.com · juansebastianpena.dev · linkedin.com/in/juan-sebastian-pena-donneys · github.com/JSebastianIEU

SUMMARY

Software engineer who builds and operates AI products end-to-end. Sole engineer on a production LLM agent that quotes and charges real customers, and technical founder of an anti-fraud ticketing platform. I work with measurable evaluation — metrics, temporal splits and tests, not just demos.

EXPERIENCE

AI Software Engineer Intern **Madrid · Remote**
Strategos AI **Feb 2026 – Present**

- Sole engineer on Craig, a production AI agent for a Dublin print shop that quotes, charges and manages customer orders end-to-end.
- Design and operate LLM multi-agent workflows against real business processes. Direct report to the CEO; the founders are non-technical.
- Own the whole stack: agent design, API, data model, deployment and operation. Continuously in production since February 2026.

Technical Founder **Colombia**
Boletapp **2026 – Present**

- Anti-fraud ticketing platform (built, in testing, pre-launch): constantly rotating HMAC-derived QR codes and an offline-first door scanner that validates without connectivity.
- One cryptographic engine implemented twice, in TypeScript and Python, held bit-for-bit identical by shared test vectors.
- The problem it attacks: a ticket forwarded over WhatsApp stops working, because the code rotates on its own and opens the door exactly once.

GenAI Mavericks — hackathon (1 week) **Madrid**
Accenture **2025**

- First edition, 100+ students from 20 universities: a RAG solution turning legal documents into plain language (Justice 2030 Spain initiative).

SELECTED PROJECTS

TikTok Recommendation Platform
Team of 7 · Jan–Apr 2026 · Python, FastAPI, React github.com/PredictiveSocialMedia/Tik-Tok-Recommendation-System

- 98 of 282 commits, 22 pull requests opened and 9 of my teammates' PRs reviewed.
- Owned evaluation: NDCG@k ranking metrics, temporal splits respecting point-in-time boundaries, and 25 regression tests validating the metrics against analytically known ground truth.
- Built the video analysis pipeline (8 parallel branches: Whisper, OCR, captioning, keywords); cut it from over 5 minutes per video to under 60 seconds.

Semantic Engagement on TikTok — published in Towards AI
2026 · Python, transformers, FAISS github.com/JSebastianIEU/tiktok-semantic-engagement

- Do semantically similar captions perform similarly? Across 22,647 posts, 12 Mann-Whitney U tests show 10.5–13% lower engagement variance than random baselines.

Visual Place Recognition — IE Tower
2026 · Python, DINOv2, FAISS, PyTorch github.com/JSebastianIEU/ie-tower-visual-place-recognition

- Given a photo of any indoor spot in the building, predicts which floor it is on, using image embeddings and nearest-neighbour retrieval over a FAISS index.

QR Forge
Open source · 2026 · FastAPI, React, Docker, Azure github.com/JSebastianIEU/qr_forge

- 136 tests at 80% line coverage behind a CI gate; three-stage non-root Docker image; CI/CD to Azure with Trivy, Bandit and Safety scanning.

EDUCATION

IE University **Madrid, Spain**
BSc in Computer Science and Artificial Intelligence **Expected: Jul 2027**

- CUBICO Young Talented Leaders Scholarship (academic and leadership excellence). Machine Learning: 9.5/10.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C
AI & ML: LLMs and multi-agent systems, RAG, transformers and embeddings, FAISS, PyTorch, scikit-learn, computer vision
Backend & data: FastAPI, PostgreSQL, SQLAlchemy, Alembic, pandas
Infrastructure & frontend: Docker, GitHub Actions, Azure, GCP, Prometheus, React, Next.js, Tailwind

LANGUAGES

Spanish (native) · **English** (bilingual, daily professional use)